Dependency Length, Syntactic Complexity & Memory:

A Reading Time Benchmark for Sentence Processing Modeling

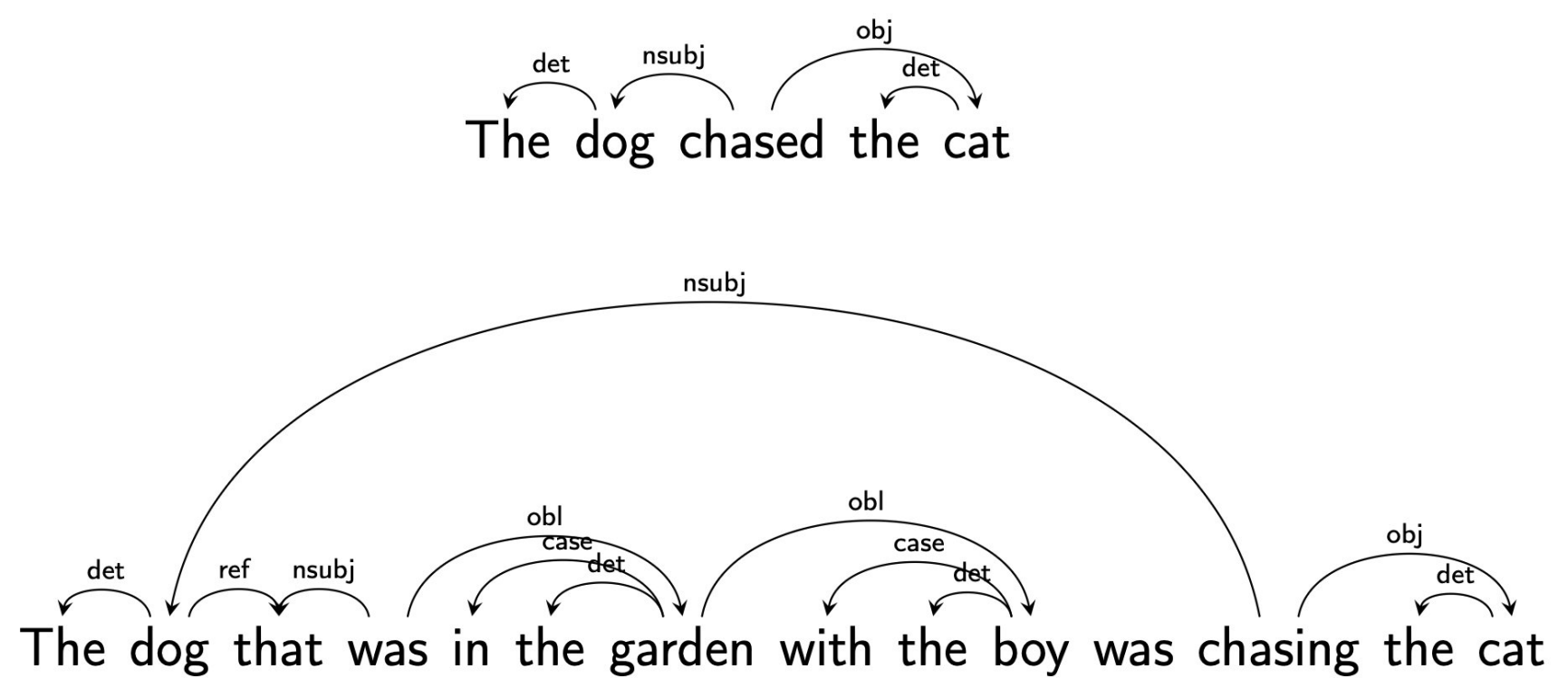
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Memory Constraints as a Critical Testbed for Processing Difficulty

Objective: To build a resource for evaluating the cognitive plausibility of language models as models of human sentence processing.

Longer dependencies → Harder memory retrieval → Higher processing cost



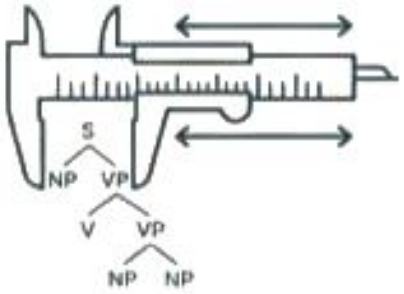
Creation of a Large, Controlled Dataset to Isolate Dependency Resolution Effects

LARGE SCALE



- 480 Native English Speakers
- 360 Unique Sentences
- 28,500 Total Reading Time Datapoints

CONTROLLED COMPLEXITY



- Systematic variation of subject-verb dependency length (0, 4, 9 words).
- Controlled syntactic complexity with up to 2 levels of embedding.

METHODOLOGY

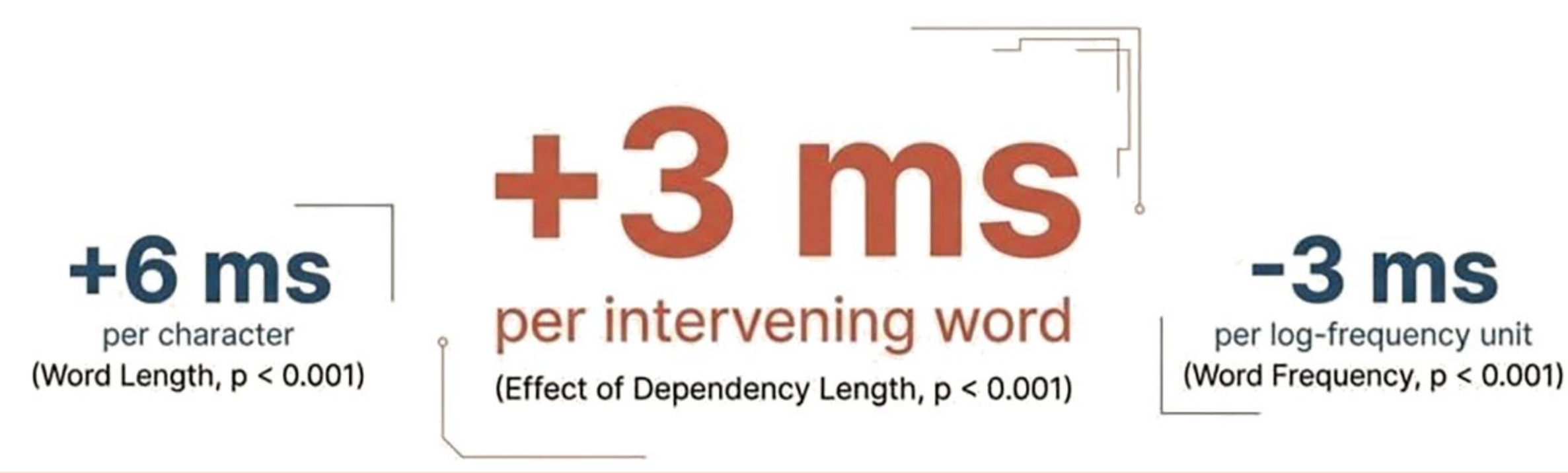


- Self-paced reading task.
- Includes comprehension questions and working memory measures.

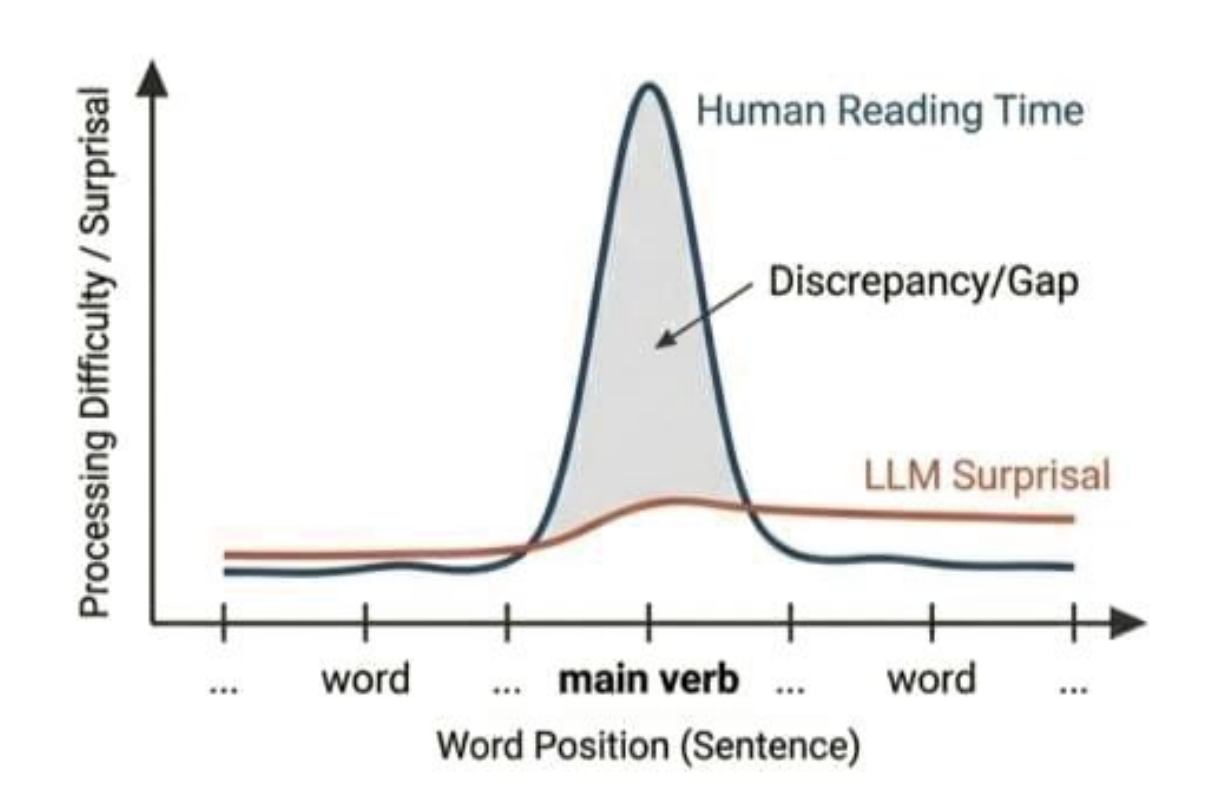
Stimuli Systematically Vary Syntactic Complexity and Dependency Length

Clause Type	Dep Length (words)	Example Set
Baseline	0	The violinist leaves the stage before the audience loudly applauds the great performance.
PP	4	The violinist in the large orchestra leaves the stage before the audience loudly applauds.
SRC	4	The violinist that followed the conductor leaves the stage before the audience loudly applauds.
ORC	4	The violinist that the conductor followed leaves the stage before the audience loudly applauds.
2×SRC	9	The violinist that followed the conductor that worked in the orchestra leaves the stage.
2×ORC	9	The violinist that the conductor that the whole orchestra admired followed leaves the stage.

Data Confirms the Hypothesis: Longer Dependencies Lead to Longer Reading Times



LLM Surprisal Underpredicts the Processing Difficulty Caused by Long Dependencies



LLM-derived surprisal does not capture the increase in processing difficulty that humans experience at the critical verb in long-dependency sentences.